

Safety-Constrained Transparent Evaluation of Levee Seepage Stability: An Ordinal Physical Floor for Learned Grades and Evidence-State Control of Inspection Data

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Abstract—Levee seepage assessments increasingly combine physics-based rule chains, data-driven classifiers and unmanned aerial vehicle (UAV) inspection, yet a single safety grade hides whether a learned model has lowered a physically identified risk or whether an inspection candidate was ever confirmed. We formulate transparent evaluation as an auditable decision chain with explicit evidence states, an ordinal physical floor that lets a random forest (RF) raise but never lower the rule-chain grade, and a multi-objective evaluation system that selects decision policies under declared safety constraints. On 875 controlled synthetic scenarios with grouped out-of-fold validation, every learner without rule input issued at least 50 direct C→A downgrades, because the learned features assign the through-structure scenarios one identical vector at every severity. The floor removed these downgrades for every learner; with the RF it corrected 71 of the rule chain’s 136 errors without introducing any and kept 0.921 ± 0.004 accuracy across five additional generator seeds. Among 4051 policies per batch, the evaluation system selected a stacked RF that takes the rule grade as input and reached 0.995 accuracy only by lowering 50 rule grades in seepage-gated scenarios. Adding the physical-floor constraint selected the stacked RF with the ordinal floor, which transferred to the five seeds at 0.939 ± 0.001 accuracy without a critical-class underestimate. All nine reviewed infrared piping candidates were false positives (exact one-sided 95% upper bound of 0.28 on candidate precision). The results hold within one synthetic generator; field labels are required before claiming field accuracy or the safety of learned downward overrides.

Index Terms—Levee seepage, piping, transparent evaluation, physics-guided machine learning, random forest, multi-objective decision, Pareto analysis, ordinal safety grade, UAV infrared inspection, evidence provenance.

I. INTRODUCTION

INTERNAL erosion and overtopping dominate the historical failure record of embankment structures: in a review of large embankment dams, piping through the embankment, the foundation or their interface accounted for close to half of the failures and overtopping for a similar share [1], and backward erosion piping remains a governing mechanism for levees on sandy foundations [2]. Operators do not act on these mechanisms directly. They act on a safety grade that compresses soil stratigraphy, hydraulic boundary conditions,

local gradients, slope stability and visible or thermal signs of abnormal discharge into one ordinal class.

Two developments make that compression harder to audit. First, levee digital twins now combine three-dimensional geology, numerical seepage fields and airborne inspection in one scene [3], [4], so a grade can be produced from evidence that is stale, spatially misregistered or merely suspected. Second, data-driven classifiers are increasingly used beside or instead of rule-based indices [5], [6]. A classifier can improve aggregate accuracy while changing the direction of its errors, and in an ordinal safety grade the direction matters: issuing the lowest-risk grade A to a critical case C is not equivalent to issuing B to a safe case.

Existing work addresses parts of this problem. Reliability methods update piping probabilities with field performance observations [7]–[9]; evidence theory combines partial support and ignorance [6], [10]; theory-guided and interpretable machine learning argue that scientific knowledge should constrain learned models in high-stakes settings [11], [12]; and cost-sensitive learning and reject options trade errors against costs or abstention [13]–[15]. What remains unresolved is how a learned grade should be *composed* with a physics-based grade so that the composition cannot produce a false-safe downgrade, what structural property determines whether the composition helps, and how unconfirmed inspection evidence should enter that composition.

This paper asks three questions. (Q1) Under leakage-controlled validation, does a nonlinear learned comparator, here a random forest (RF), improve on a physics-based rule chain, and in which direction do its errors move? (Q2) Can a monotone physical lower bound remove false-safe errors while retaining the learned improvements, and when does that guarantee hold? (Q3) How should a decision policy be selected among learners, thresholds and floor scopes when quality, critical-class recall, safety loss and escalation burden conflict? (Q4) How should algorithmic inspection candidates and incomplete engineering evidence be represented so that they cannot silently change a grade? Our contributions are as follows.

- An evidence-state and admissibility model that separates missing, inapplicable, candidate, reviewed and admissible evidence and records the provenance of every value that reaches a grade (Section III).

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Code, data, per-record predictions and figure sources are in the accompanying de-identified repository (Section VIII).

- A leakage-controlled benchmark protocol with error-direction metrics, and an analysis showing that the false-safe errors of the learned layer arise from a feature-set blind spot that no threshold choice can remove (Sections V and VI-B).
- An ordinal physical floor, a proposition that characterizes its error sets, and its evaluation on the original batch, five additional generator seeds, a floor-scope ablation and a review-budget sweep (Sections III-D, VI-C and VI-D).
- A multi-objective evaluation system with offline and online decision algorithms, declared constraints and a hypervolume comparison of policy families, evaluated against eight comparison baselines that include learned-only, cost-sensitive and rule-aware fusion methods (Sections IV, VI-F and VI-G).
- A field audit that treats unmanned aerial vehicle (UAV) infrared detections and a delivered engineering case as evidence-state and provenance tests rather than as accuracy tests (Sections VI-H and VI-I).

The controlled benchmark and the additional seeds come from one synthetic generator. The results therefore establish the behaviour and the failure conditions of the decision protocol, not field-wide predictive accuracy.

II. RELATED WORK

A. Physical and numerical evidence for seepage safety

Implicit geological modelling interpolates stratigraphic interfaces from boreholes and sections, and the interpolation method and conditioning data are inseparable from the resulting geometry [16], [17]. Transient seepage is commonly described by Richards-type formulations [18], [19] with unsaturated properties such as the van Genuchten relationship [20], and piping models relate local gradients at the exit to erosion progression [2]. Reliability and Bayesian updating propagate parameter variability and incorporate proven-strength observations [5], [7]–[9]. These methods produce physical responses and probabilities, but they do not specify when a response extracted from a particular material zone, scenario and version is admissible input to an operational grade. That admissibility rule is one of the gaps addressed here.

B. Digital twins and provenance

Digital-twin studies describe a loop linking sensing, model updating, simulation and decision support [3], and building information modelling has been combined with mechanism algorithms for the full life cycle of levees [4]. Provenance standards such as PROV-DM model how entities derive from activities and agents [21]. Synchronizing a virtual levee with its physical counterpart does not, however, decide whether a stale, unregistered or merely suspected observation may constrain a safety decision. We therefore use the twin as context and define admissibility separately.

C. Learned safety classifiers and physical guidance

Data-driven seepage evaluation has used surrogate classifiers for reliability analysis [5] and deep learning combined

with Dempster–Shafer fusion for dam seepage [6]. Tree ensembles [22]–[24] are attractive comparators because they model interactions among heterogeneous indicators and admit feature-level inspection [25]. Theory-guided data science and the case for interpretable models in high-stakes decisions both argue that scientific knowledge should constrain the learner [11], [12]. Most such approaches constrain the model’s inputs, loss or structure. Few constrain the *direction* of its errors at the decision layer, which is what an ordinal safety grade with asymmetric error costs requires [13].

D. Inspection candidates and selective decisions

UAV photogrammetry and thermography can cover long levee reaches, provided that georegistration, radiometric calibration and flight metadata support the intended inference [26]. Segmentation and detection networks [27], [28] and thermographic deep-learning frameworks [29] can localize seepage-like signatures, but water boundaries, vegetation, shadow and wet ground produce similar thermal contrast. Selective classification and reject options formalize abstention when confidence is low [14], [15]. In levee practice, abstention means sending a record to an engineer, so the review rate and the consequences of the fallback must be measured rather than assumed.

III. TRANSPARENT EVALUATION FRAMEWORK

Fig. 1 summarizes the decision chain. Evidence passes an admissibility gate, the physics-based rule chain and a learned layer each produce a grade, an ordinal physical floor combines them, and an optional review gate routes low-confidence records to an engineer. Every output carries the triggered gates, the evidence states and the version identifiers used to produce it.

A. Evaluation unit, evidence states and admissibility

An evaluation record is indexed by (s, w, t, v) : levee section s , hydraulic scenario w , observation or simulation time t , and evidence-and-rule version v . Each indicator retains its raw value, unit, transformed score, source location, value type (direct measurement, numerical output, representative assignment, controlled synthetic value or manual assignment), applicability, availability, time window, spatial support and transformation rule.

Applicability and availability are separate fields. An indicator is *inapplicable* only after the corresponding facility or failure mode has been confirmed absent; it is *missing* when it should be evaluated but evidence was not obtained. Neither state is encoded as a zero or as a perfect score. Inspection evidence has three further states: *candidate* (algorithmic detection), *reviewed* (manual status recorded) and *admissible* (time, location and status closed for the current unit). Only an admissible confirmed event can impose a deterministic piping or defect constraint. Table I lists the channels and their admissibility conditions.

The evidence table is implemented as a directed provenance graph in the sense of PROV-DM [21]. Nodes represent observations, geological objects, numerical fields, review actions

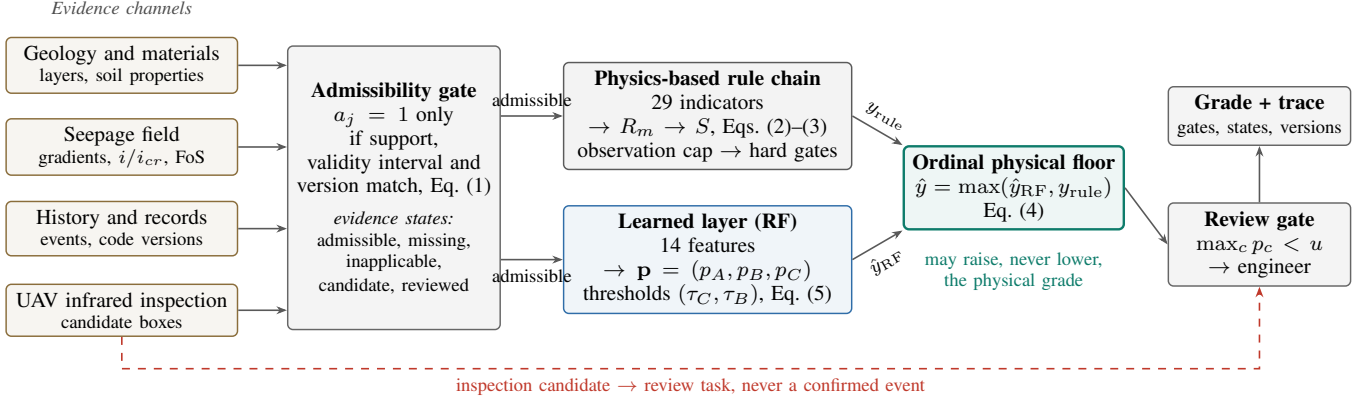


Fig. 1. Transparent evaluation chain (FoS, factor of safety). Evidence enters only through the admissibility gate, which records an explicit state for every indicator. The rule chain and the learned layer each return an ordinal grade (A < B < C in risk). The ordinal physical floor lets the learned layer escalate, but not downgrade, the rule grade. Inspection candidates create review tasks; they constrain a grade only after review closes their time, location and status.

TABLE I
EVIDENCE CHANNELS AND ADMISSIBILITY CONDITIONS.

Channel	Example evidence	Admissibility condition
Geology	control layer, soil properties, contact zone	spatially registered; material identity retained in the numerical domain
Seepage field	local gradient, i/i_{cr} , factor of safety	scenario, boundary condition, extraction region and material known
Inspection	candidate box, confidence, review status	capture time, location and review state stored; candidate $\neq$ confirmed
History	previous piping or seepage event	event interval distinguished from current relevance
Missingness	unavailable or stale input	reported as a state with a missing-item list

and decision outputs; edges record derivation, spatial registration, temporal precedence and rule triggering. For a unit (s, w, t, v) , indicator x_j is admissible only when its spatial support $\mathcal{S}_j$ intersects the unit support $\mathcal{S}_s$, its validity interval $\mathcal{T}_j$ contains t , and its version v_j is compatible with v :

$$a_j(s, w, t, v) = \mathbf{1}\{\mathcal{S}_j \cap \mathcal{S}_s \neq \emptyset\} \mathbf{1}\{t \in \mathcal{T}_j\} \mathbf{1}\{v_j \preceq v\}. \quad (1)$$

If $a_j = 0$, the value is retained for traceability but cannot become a normal score. A misregistered inspection box or a stale simulation output therefore appears as a provenance failure rather than as a numerical zero.

B. Physics-based rule chain

The rule chain contains 29 secondary indicators and five failure modes: overtopping, seepage, slope instability, scour and through-structure effects. Let $R_m \in [0, 1]$ be the management-barrier-adjusted response of failure mode m , π_m its normalized weight and α a shortfall coefficient. The composite risk measure and the score are

$$\rho = \alpha \max_m R_m + (1 - \alpha) \sum_m \pi_m R_m, \quad (2)$$

$$S = 100(1 - \rho), \quad (3)$$

with $\alpha = 0.45$. The score grade is A for $S \geq 80$, B for $40 \leq S < 80$ and C for $S < 40$. An observation cap and physical hard gates are then applied in sequence; the gates cover overtopping, the seepage indicators C1, C4 and C5 (foundation, embankment and contact zone), slope factors of safety, scour, structures and settlement. Each stage may keep the grade or move it towards higher risk, never towards lower risk. The dominant failure mode is the largest internal response and is not a calibrated failure probability.

C. Random-forest decision layer

The learned layer uses 14 features from the physical and inspection channels: crest elevation; foundation, embankment and contact-zone gradients; downstream and upstream slope factors of safety; scour depth; inspection anomaly density and evidence level; the three gradients divided by 0.25; the inverse downstream factor of safety; and the scour depth divided by a 1.0 m reference, which duplicates the raw depth. It never sees target mode, target severity, any delivered grade, the composite score, the risk measure or the dominant mode. The 14 features contain no structure-condition indicator; Section VI-B shows the consequence.

Both learners are deterministic NumPy implementations of classification and regression trees (CART) with Gini impurity [22], [24]. Candidate split thresholds are the midpoints between consecutive distinct feature values; when a node holds more than 30 distinct values, the midpoints are taken between 19 quantiles from the 5th to the 95th percentile. The decision tree uses maximum depth 5 and minimum leaf size 5 with all features. The random forest [23] uses 120 bootstrap trees, maximum depth 7, minimum leaf size 3 and four randomly drawn features per split; its class probability is the fraction of trees voting for the class. Cost-weighted variants draw bootstrap samples with probability proportional to a class weight $w_C \in \{1.5, 2, 3, 4, 6\}$ for class C [13]. All hyperparameters were fixed before the additional seeds were generated and were not retuned. The probability vector $\mathbf{p} = (p_A, p_B, p_C)$ is a decision-support score under the controlled target construction, not a calibrated field failure probability. The RF is the default

learner of the decision layer; Section IV-E adds four further learners, an expected-cost decision rule and three fusion rules for comparison.

D. Ordinal physical floor

Encode A, B and C as 0, 1 and 2. Given the learned decision $\hat{y}_{\text{RF}}$ and the final grade y_{rule} of the rule chain, the ordinal physical floor returns

$$\hat{y}_{\text{H}} = \max(\hat{y}_{\text{RF}}, y_{\text{rule}}). \quad (4)$$

We call this policy the *floor hybrid*; tables and figures label it RF + ordinal floor. It retrain neither component and leaves the evidence trace and triggered gates intact. Its error behaviour follows directly from the order structure.

Proposition 1: For a record with reference class y , (i) $\hat{y}_{\text{H}} < y$ if and only if $\hat{y}_{\text{RF}} < y$ and $y_{\text{rule}} < y$; (ii) $\hat{y}_{\text{H}} > y$ if and only if $\hat{y}_{\text{RF}} > y$ or $y_{\text{rule}} > y$; (iii) $\hat{y}_{\text{H}} = 0$ if and only if both components return 0. Consequently, the hybrid's underestimates are the intersection of the components' underestimates, its overestimates are their union, and the hybrid issues a C→A decision only where the rule chain does.

Proof: For integers a, b, y : $\max(a, b) < y \Leftrightarrow a < y \wedge b < y$; $\max(a, b) > y \Leftrightarrow a > y \vee b > y$; and $\max(a, b) = 0 \Leftrightarrow a = b = 0$ for $a, b \geq 0$. ■

Proposition 1 states when the floor helps and when it fails. The hybrid gains when the two components err on different records and the learned layer rarely overestimates; it inherits every conservative error of the rule chain; and its protection against C→A decisions is exactly as good as the rule chain's. It is not a statement about the accuracy of either component. We also evaluate a *C-only floor* (RF + C-only floor), $\hat{y} = 2$ if $y_{\text{rule}} = 2$ and $\hat{y} = \hat{y}_{\text{RF}}$ otherwise, which removes the rule chain's influence below C.

IV. DECISION ALGORITHM AND MULTI-OBJECTIVE EVALUATION SYSTEM

A single declared operating point answers whether the floor works; it does not show how the decision layer should be configured when quality, critical-class recall, safety loss and escalation burden conflict, or whether another learner or fusion rule would serve better. This section defines the policy space, the objectives and constraints, the selection rule and the decision algorithms, and the comparison baselines. Fig. 2 summarizes the evaluation system.

A. Decision policies

A decision policy is a tuple $\pi = (\ell, \tau_C, \tau_B, \phi)$ of a learner ℓ , a threshold pair and a floor scope $\phi \in \{\text{none}, \text{C-only}, \text{ordinal}\}$. The learner returns OOF class probabilities $\mathbf{p}^\ell = (p_A, p_B, p_C)$, which the thresholds map to a learned grade

$$\hat{y}_\ell = \begin{cases} C, & p_C \geq \tau_C, \\ B, & p_C < \tau_C \text{ and } p_B \geq \tau_B, \\ A, & \text{otherwise,} \end{cases} \quad (5)$$

and the floor scope combines it with the rule grade:

$$\hat{y}_\pi = \begin{cases} \hat{y}_\ell, & \phi = \text{none}, \\ C \text{ if } y_{\text{rule}} = C, \text{ else } \hat{y}_\ell, & \phi = \text{C-only}, \\ \max(\hat{y}_\ell, y_{\text{rule}}), & \phi = \text{ordinal}. \end{cases} \quad (6)$$

The ordinal scope is the floor of Eq. (4), and Proposition 1 applies to it for any learner. The thresholds range over the 0.05 grid $G = \{0.20, 0.25, \dots, 0.90\}^2$ (225 pairs), so each batch contains $6 \times 225 \times 3$ learned policies and the rule chain itself, 4051 in total. The representative RF operating point $(\tau_C, \tau_B) = (0.35, 0.20)$ used in Sections VI-A–VI-E was declared on this grid as the highest macro-F1 point with zero A-class overestimates and C recall of at least 0.85; it is a declared reference, not an optimum. A 0.01 grid of 10,000 RF pairs tests whether any threshold alone avoids C→A errors. A review gate with threshold u can be attached to any policy (Algorithm 2); because no reviewer decisions exist offline, review is evaluated separately with the rule grade as a proxy (Section VI-E).

B. Objectives, resources and constraints

Because accuracy alone is uninformative when classes are imbalanced and errors are asymmetric [30], each policy is described by four objectives,

$$\mathbf{F}(\pi) = (\text{mF1}, \text{BA}, R_C, 1 - \frac{L_{\text{safe}}}{L_{\text{max}}}) \in [0, 1]^4, \quad (7)$$

all to be maximized: macro-F1, balanced accuracy, C recall, and the normalized safety loss

$$L_{\text{safe}} = 2N(C \rightarrow A) + N(C \rightarrow B), \quad L_{\text{max}} = 2N_C, \quad (8)$$

which weights a direct two-level downgrade twice as heavily as a one-level downgrade. Policy π dominates π' when it is at least as good in every objective and strictly better in at least one,

$$\pi \succ \pi' \iff \mathbf{F}(\pi) \geq \mathbf{F}(\pi') \wedge \mathbf{F}(\pi) \neq \mathbf{F}(\pi'), \quad (9)$$

and the Pareto set contains the non-dominated policies [31]. Three resource coordinates are reported but kept out of the dominance relation: the escalation rate (fraction of units assigned C), the number of A-class overestimates, and the review rate. Treating them as objectives would let a policy appear efficient by escalating less while missing more critical cases.

Two constraint sets are declared before evaluation. The benchmark constraints admit policies with $N(C \rightarrow A) = 0$ and $R_C \geq 0.95$. The physical-floor constraint additionally requires $\hat{y}_\pi \geq y_{\text{rule}}$ on every unit, that is, no learned decision may lower a physics-based grade. The benchmark can measure whether a downward override is correct under its own reference labels, but not whether it is safe on a real levee, so the second constraint encodes the design principle of Section III-D rather than a measured cost.

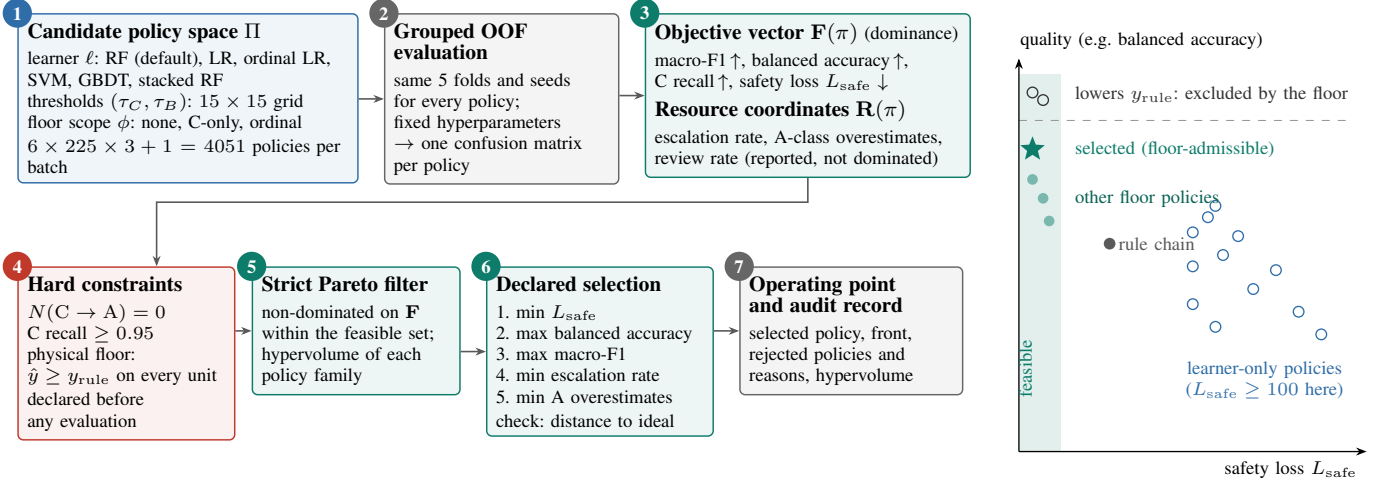


Fig. 2. Multi-objective evaluation system. (1) Candidate policies combine a learner, a threshold pair and a floor scope. (2) Every policy is scored on the same grouped OOF predictions. (3) Four objectives define dominance; resource coordinates are reported but not traded against safety. (4) Hard constraints, including the optional physical-floor constraint, are declared before evaluation. (5) The strict Pareto filter retains the non-dominated feasible policies, and hypervolume summarizes each policy family. (6) A declared lexicographic rule selects one operating point, with a distance-to-ideal rule as a sensitivity check. (7) The selected point is released with its audit record. The inset sketches the geometry found in Section VI-G: learner-only policies sit at high safety loss, and the best benchmark policies lower the rule grade and are excluded by the physical floor.

C. Selection rule and hypervolume

Within the feasible Pareto set, the operating point is selected by a declared safety-first lexicographic order: minimum L_{safe} , then maximum balanced accuracy, then maximum macro-F1, then minimum escalation rate, then minimum A-class overestimates. As a sensitivity check, the policy with the smallest equal-weight Euclidean distance to the ideal point of the min-max normalized objectives is also reported. Each policy family $\mathcal{F} = (\ell, \phi)$ is summarized by the hypervolume of its non-dominated set [32], [33],

$$\text{HV}(\mathcal{F}) = \lambda\left(\bigcup_{\pi \in \mathcal{F}} [\mathbf{0}, \mathbf{F}(\pi)]\right), \quad (10)$$

where λ is the Lebesgue measure on $[0, 1]^4$ and the reference point is the origin. A larger hypervolume means that the family reaches a larger part of the objective space; the rule chain's own hypervolume serves as the physics-only reference. Because selection and evaluation on the same batch are optimistic, the policy selected on the original batch is also frozen and evaluated unchanged on the five additional seeds (transfer test).

D. Decision algorithms

Algorithm 1 calibrates the decision layer offline on a reference batch and records the evidence for the chosen operating point. Algorithm 2 grades a new evaluation unit with the frozen policy. The online procedure never computes the safety loss, which requires a reference class, and records instead whether and by how many levels the floor raised the learned grade.

E. Comparison baselines

All baselines use the same 14 features, folds and reference labels as the RF, and none was tuned; library defaults or the fixed settings below were set before any result was seen.

Algorithm 1 Offline calibration and policy selection

Require: reference batch $\{(\mathbf{x}_i, y_i, y_{\text{rule}, i})\}$; learners $\mathcal{L}$; grid G ; scopes Φ ; constraint set $\mathcal{C}$

Ensure: operating point π^* and audit record

- 1: group identical feature vectors; assign groups to $K = 5$ folds
- 2: **for** each $\ell \in \mathcal{L}$ and fold k **do**
- 3: fit ℓ on folds $\neq k$; store $\mathbf{p}_i^\ell$ for i in fold k
- 4: **end for**
- 5: **for** each $\pi = (\ell, \tau_C, \tau_B, \phi)$ with $(\tau_C, \tau_B) \in G, \phi \in \Phi$ **do**
- 6: $\hat{y}_{\pi, i} \leftarrow$ Eqs. (5)–(6); compute $\mathbf{F}(\pi)$, resources, $N(C \rightarrow A)$, downward overrides
- 7: **end for**
- 8: $\Pi_{\mathcal{C}} \leftarrow \{\pi : \pi \text{ satisfies } \mathcal{C}\}$; $P \leftarrow$ non-dominated set of $\Pi_{\mathcal{C}}$ (Eq. 9)
- 9: $\text{HV}(\ell, \phi) \leftarrow$ Eq. (10) for every family
- 10: $\pi^* \leftarrow$ first element of P under the lexicographic order
- 11: **return** π^* ; record P , HV, π^* , rejected policies and the violated constraint

Learned-only baselines are multinomial logistic regression (LR) on standardized features ($C = 1$); an ordinal LR that decomposes the grade into $P(y > A)$ and $P(y > B)$ [34]; a radial-basis support-vector machine (SVM, $C = 1$) with sigmoid-calibrated probabilities [35]; gradient-boosted trees (GBDT, 200 iterations, learning rate 0.1) [36]; and the RF with an expected-cost decision $\arg \min_k \sum_c p_c \kappa_{ck}$, whose cost matrix κ charges 10 for $C \rightarrow A$, 5 for $C \rightarrow B$, 2 for $B \rightarrow A$ and 1 or 2 for overestimates [13]. The sklearn implementations [37] (version 1.9.1) were used for LR, SVM and GBDT. *Rule-aware fusion* baselines use the rule grade at decision time: a soft vote that averages $\mathbf{p}^{\text{RF}}$ with the one-hot rule grade and resolves ties towards higher risk; a confidence switch that

Algorithm 2 Online grading of an evaluation unit

Require: unit key (s, w, t, v) ; evidence set E ; frozen $\pi^* = (\ell, \tau_C, \tau_B, \phi)$; review threshold u

Ensure: grade, trace and review status

- 1: freeze (s, w, t, v) ; for each indicator j compute a_j (Eq. 1) and its evidence state
 - 2: keep inadmissible values in the trace only; list missing indicators
 - 3: $y_{\text{rule}}, \text{triggered gates} \leftarrow$ rule chain on admissible evidence (Eqs. 2–3)
 - 4: $\mathbf{p} \leftarrow \ell(\mathbf{x})$; $\hat{y}_\ell \leftarrow$ Eq. (5); $\hat{y} \leftarrow$ Eq. (6); record $\hat{y} - \hat{y}_\ell$
 - 5: **if** $\max_c p_c < u$ **or** a candidate inspection state touches the unit **then**
 - 6: create a review task; mark the grade provisional
 - 7: **end if**
 - 8: publish $\hat{y}$ with gates, evidence states, provenance, versions and review status
-

TABLE II
VALIDATION STATUS OF THE DECISION-LAYER COMPONENTS.

Component	Evidence in this paper	Status
Ordinal floor, Eq. (4)	Proposition 1; six controlled batches	guarantee proven, conditional on the rule chain
RF and baseline learners	grouped OOF, six batches	within-generator only
Objectives, Eq. (7); 2:1 weight in L_{safe}	declared	not calibrated to consequences
Constraints: $N(C \rightarrow A) = 0$, $R_C \geq 0.95$, physical floor	declared; selection reported with and without the floor	thresholds need field calibration
Lexicographic selection	distance-to-ideal check; transfer to five seeds	within-generator only
Review gate, u and 5% budget	rule-grade proxy	no reviewer decisions
Evidence states, Eq. (1)	inspection package; K45+500 trace	interface tested, not outcome-validated
Learned downward overrides	correct on benchmark labels	untested on field labels

uses the RF threshold decision when $\max_c p_c \geq 0.60$ and the rule grade otherwise; and a stacked RF that receives the rule grade as a fifteenth feature [38]. Every learned-only baseline and the stacked RF are also evaluated with the ordinal floor. Unless stated otherwise, learned-only baselines use the argmax decision.

F. Validation status of the decision layer

Table II separates what the present data establish from what still requires field evidence. Components marked as declared are design choices that were fixed before evaluation but have not been calibrated against consequences, costs or field outcomes.

V. DATA AND VALIDATION PROTOCOL

A. Controlled benchmark

The benchmark was produced by a frozen generator and contains 875 records: seven target modes (safe reference, overtopping, seepage, slope, scour, through-structure and mixed)

$\times$ five severities (safe, slight, moderate, severe, critical) $\times$ 25 records. The reference class follows the frozen severity mapping: safe and slight $\rightarrow$ A (350 records), moderate $\rightarrow$ B (175), severe and critical $\rightarrow$ C (350). It is a closed-set reference for testing decision behaviour under the generator, not a field safety label. The rule chain’s final grades were reconstructed from the delivered final-grade error trace rather than from the raw grade column of the input file, and a replay of the generator reproduced all five delivered benchmark files byte for byte.

B. Splits and validation hierarchy

The 875 records contain 722 distinct 14-feature vectors. Identical vectors are assigned to the same fold, and all learned results are grouped five-fold out-of-fold (OOF) predictions, which prevents duplicated inputs from leaking across folds [39], [40]. Forest seeds are derived from the fold index and held fixed across methods. A five-outer by three-inner grouped nested validation re-selects (τ_C, τ_B) inside each outer fold (with 80 trees). Validation is organized in three levels: the original 875 records test behaviour against the frozen reference; five additional 875-record batches (generator seeds 20260930–20260934, same construction, features, splits, hyperparameters and policies) test sensitivity to the generator realization; and the inspection and engineering packages test whether evidence states and provenance survive the interface. None of the latter is used to claim external predictive accuracy.

C. Statistical analysis

Paired comparisons with the rule chain on the same records use the exact two-sided McNemar test on per-record correctness [41], [42]. Within-batch uncertainty of the hybrid-minus-rule differences is summarized by a paired percentile bootstrap that resamples the 722 feature-vector groups (2000 resamples) [43]; it conditions on the fitted OOF predictions and does not include retraining variability. Results across the five additional seeds are reported as mean $\pm$ sample standard deviation (SD) without inferential claims, because five realizations of one generator do not support population inference. For the 0/9 confirmed field candidates we report the exact Clopper–Pearson bound [44].

D. Field inspection package

The inspection package contains 114 field thermal infrared images with GPS and capture-time metadata, nine algorithmic piping candidates from those images that entered manual review, and 13 images of artificial piping with 14 detector boxes. The remaining field images have no image-level truth labels, and the artificial set has no record of missed detections, so neither supports a detection accuracy, recall or F1 estimate.

E. Engineering-case trace

The delivered K45+500 case, a representative section of a reach with a recorded 2017 piping event, is used only to test provenance and interface fields. Its values are delivery excerpts rather than an independent physical re-run, and its input includes the historical piping information, so it is not a blind test.

TABLE III

GROUPED FIVE-FOLD OOF COMPARISON ON THE 875 CONTROLLED RECORDS. $C \rightarrow A$ AND $C \rightarrow B$ COUNT C-CLASS UNDERESTIMATES (350 C RECORDS); $A \uparrow$ COUNTS A-CLASS OVERESTIMATES (350 A RECORDS); ESC. IS THE FRACTION OF RECORDS ASSIGNED C (REFERENCE PREVALENCE 0.400). THE PROPOSED ORDINAL FLOOR IS SHADED.

	Method	Accuracy	Macro-F1	Bal. acc.	Recall C	$C \rightarrow A$	$C \rightarrow B$	L_{safe}	$A \uparrow$	Esc.
Physics	Rule chain	.845	.827	.847	.837	0	57	57	54	.335
	Decision tree	.707	.646	.635	.631	129	0	258	0	.253
Learned only	RF, majority vote	.829	.784	.763	.854	51	0	102	0	.342
	RF, threshold (0.35, 0.20)	.873	.858	.836	.857	50	0	100	0	.343
	RF, best of 0.01 grid (0.19, 0.06)	.887	.890	.882	.857	50	0	100	24	.361
	RF, C weight 6	.707	.701	.683	1.000	0	0	0	179	.683
	RF + C-only floor	.930	.903	.884	1.000	0	0	0	0	.400
Floor policies	RF + ordinal floor (Eq. 4)	.926	.912	.928	1.000	0	0	0	54	.400
	RF + ordinal floor + review, $u = 0.60$	.911	.896	.914	.966	0	12	12	54	.386

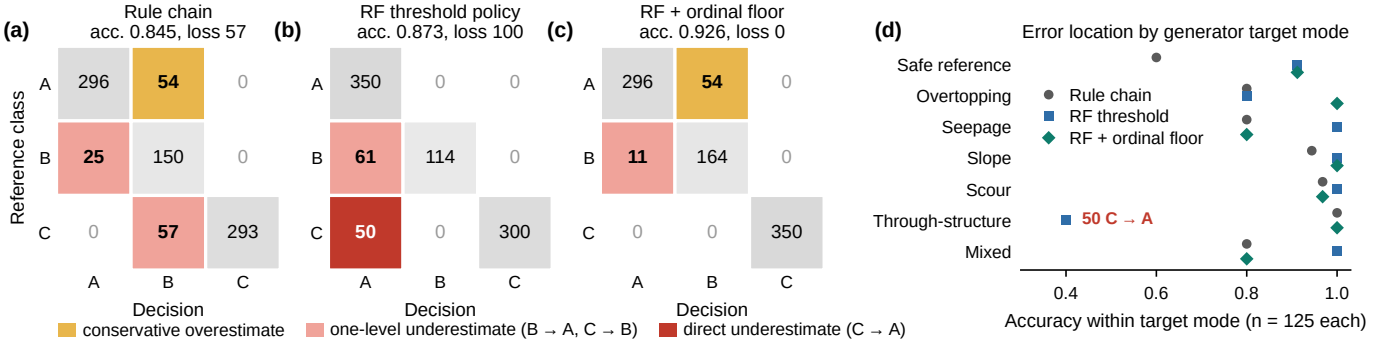


Fig. 3. Error structure on the 875 controlled records. (a)–(c) Confusion matrices for the rule chain, the RF threshold policy $(\tau_C, \tau_B) = (0.35, 0.20)$ and the floor hybrid (RF + ordinal floor); cell colour encodes error direction. The RF policy removes the rule chain’s $C \rightarrow B$ errors but converts its C-class misses into 50 direct $C \rightarrow A$ decisions. The floor hybrid keeps the rule chain’s 54 conservative $A \rightarrow B$ errors and none of the underestimates of either component. (d) Accuracy within each generator target mode ($n = 125$). All RF $C \rightarrow A$ errors occur in the through-structure mode, where every record shares one 14-feature vector.

F. Reproducibility

All analyses use fixed seeds and write their feature manifests, fold assignments, threshold grids, per-record predictions and summary metrics. The supplementary statistics, the comparison baselines, the policy-space evaluation and all data figures are regenerated by three scripts that read only repository files and reuse the unchanged model code (NumPy 2.4.6 and scikit-learn 1.9.1; Section VIII).

VI. RESULTS

A. The rule chain errs conservatively, and its gates do the work

The rule chain classified 739 of 875 records correctly (accuracy 0.845, macro-F1 0.827, balanced accuracy 0.847; Table III). All 136 errors were one-level errors (Fig. 3a): 54 $A \rightarrow B$ and 25 $B \rightarrow A$ decisions and 57 $C \rightarrow B$ underestimates, with no $C \rightarrow A$ or $A \rightarrow C$ decision. The score grade alone was correct for only 450 records; the observation cap changed no grade, and the hard gates changed 322 grades. An ablation of the inspection channel confirmed its inertness in this benchmark: replacing the inspection indicator C3 with its no-anomaly anchor changed 150 scores by at most 1.39 points, but no grade. The rule chain’s accuracy therefore rests on gates that are aligned with the generator’s severity construction,

which makes it a strong comparator and is a property of the benchmark rather than evidence of field skill.

B. Learned policies improve aggregate metrics but create false-safe downgrades

The shallow tree underperformed the rule chain on every aggregate metric (Table III). The majority-vote forest raised C recall from 0.837 to 0.854 but lowered accuracy and macro-F1, and all 51 of its C-class misses were direct $C \rightarrow A$ decisions, doubling L_{safe} from 57 to 102. The representative threshold policy reached 0.873 accuracy and 0.858 macro-F1. Its accuracy gain over the rule chain was not distinguishable within the batch (125 records correct only under the RF policy against 100 correct only under the rule chain; exact McNemar $p = 0.11$), and all 50 of its C-class misses were again $C \rightarrow A$ ($L_{\text{safe}} = 100$).

Threshold tuning cannot remove these errors. Of the 10,000 pairs on the 0.01 grid, none produced fewer than 50 $C \rightarrow A$ decisions; the best pair by macro-F1, (0.19, 0.06), reached 0.890 macro-F1 with the same 50 $C \rightarrow A$ errors (Fig. 4). Cost weighting removed them only at $w_C = 6$, where accuracy fell to 0.707 and 179 of 350 A records were escalated; weights 1.5 to 4 left 50 or 51 $C \rightarrow A$ errors. The nested validation reproduced the pattern (accuracy 0.881, macro-F1 0.870, 50 $C \rightarrow A$ errors in the pooled outer predictions).

TABLE IV

FIVE ADDITIONAL GENERATOR SEEDS (MEAN $\pm$ SAMPLE SD OVER SEEDS 20260930–20260934; 875 RECORDS EACH). REVIEW RATE IS THE FRACTION OF RECORDS WITH $\max_c p_c < u$, ROUTED TO THE RULE GRADE AS AN OFFLINE REVIEWER PROXY.

Method	Accuracy	Macro-F1	Bal. acc.	Recall C	C $\rightarrow$ A	L_{safe}	A $\uparrow$	Review (%)
Rule chain	.847 $\pm$.002	.830 $\pm$.002	.849 $\pm$.001	.842 $\pm$.002	0	55.4 $\pm$ 0.5	53.2 $\pm$ 1.1	–
RF, majority vote	.827 $\pm$.008	.789 $\pm$.013	.767 $\pm$.012	.837 $\pm$.013	57.0 $\pm$ 4.5	114.0 $\pm$ 9.1	0	–
RF, threshold (0.35, 0.20)	.875 $\pm$.010	.862 $\pm$.015	.841 $\pm$.016	.855 $\pm$.004	50.6 $\pm$ 1.3	101.2 $\pm$ 2.7	0	–
RF + C-only floor	.933 $\pm$.010	.907 $\pm$.015	.888 $\pm$.016	.998 $\pm$.004	0.6 $\pm$ 1.3	1.2 $\pm$ 2.7	0	–
RF + ordinal floor	.921 $\pm$.004	.906 $\pm$.005	.920 $\pm$.006	.998 $\pm$.004	0	0.6 $\pm$ 1.3	53.2 $\pm$ 1.1	–
RF + ordinal floor + review, $u = 0.50$	.918 $\pm$.006	.902 $\pm$.007	.917 $\pm$.007	.990 $\pm$.008	0	3.4 $\pm$ 2.8	53.2 $\pm$ 1.1	1.5 $\pm$ 0.7
RF + ordinal floor + review, $u = 0.60$	.908 $\pm$.006	.892 $\pm$.007	.908 $\pm$.008	.969 $\pm$.008	0	10.8 $\pm$ 2.8	53.2 $\pm$ 1.1	6.6 $\pm$ 0.9

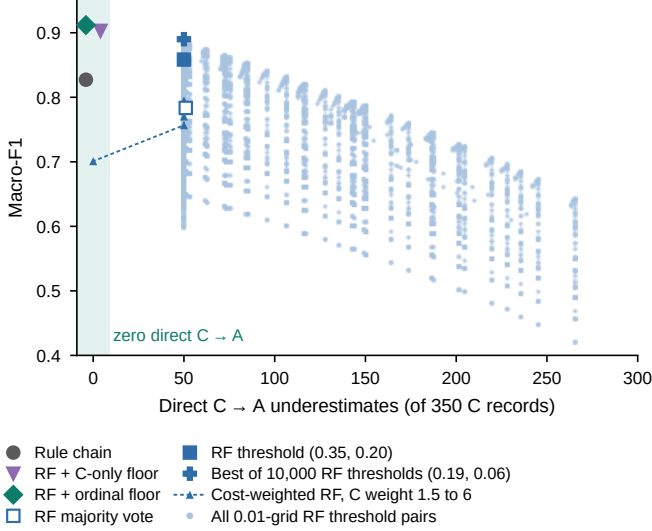


Fig. 4. Quality against direct C $\rightarrow$ A errors on the 875 controlled records. Light points are all 10,000 RF threshold pairs on the 0.01 grid; no pure RF threshold reaches fewer than 50 C $\rightarrow$ A errors. The dashed line traces cost-weighted forests ($w_C = 1.5$ to 6). Only the rule chain, the C-weight-6 forest and the two floor policies reach zero, and RF + ordinal floor has the highest macro-F1 among them. Points at zero are offset horizontally by up to four units so that none is hidden.

The cause is a blind spot in the learned features rather than a poor threshold. All 50 C $\rightarrow$ A errors occur in the through-structure target mode (Fig. 3d), whose severity is expressed through structure indicators that the rule chain gates on but the 14 learned features do not contain. In that mode all 125 records, from safe to critical, share one identical feature vector, the same vector as the safe reference scenarios. No function of those features can separate them. The learned layer is nevertheless complementary to the rule chain elsewhere: it corrected the 57 C $\rightarrow$ B errors that the rule chain made in the safe-reference, overtopping and slope modes and the 54 A $\rightarrow$ B errors in the seepage, mixed and scour modes.

C. The ordinal floor removes underestimates without adding errors

The floor hybrid reached 0.926 accuracy, 0.912 macro-F1, 0.928 balanced accuracy and C recall 1.000, with no C-class underestimate and $L_{\text{safe}} = 0$ (Table III, Fig. 3c). Relative to the rule chain it corrected 71 of 136 errors and introduced none (exact McNemar $p < 10^{-20}$). The paired

group bootstrap gave differences of +8.1 percentage points in accuracy (95% interval 5.4 to 11.5), +8.5 points in macro-F1 (6.0 to 11.3) and -57 in L_{safe} (-72 to -42). Relative to the RF threshold policy, the floor changed 154 decisions: it corrected 100 RF errors (all 50 C $\rightarrow$ A and 50 of 61 B $\rightarrow$ A) and introduced 54 A $\rightarrow$ B escalations, which are exactly the rule chain’s conservative errors.

Proposition 1 accounts for each of these counts. The rule chain and the RF policy never missed the same C record, so the intersection of their C underestimates, and hence the hybrid’s, was empty. The RF policy made no overestimate, so the union of overestimates equals the rule chain’s 54. The 11 remaining B $\rightarrow$ A errors are safe-reference records that both components placed in A. The hybrid’s escalation rate of 0.400 equals the C prevalence of the benchmark.

The scope of the floor matters. The C-only floor reached a higher accuracy (0.930) and avoided the 54 A $\rightarrow$ B escalations, but its balanced accuracy fell to 0.884 because it discarded the rule chain’s B-level corrections (61 B $\rightarrow$ A errors remained). More importantly, it does not inherit the C $\rightarrow$ A guarantee of Proposition 1(iii): a C record that the rule chain places in B and the learned layer places in A is issued A. This occurred in one of the five additional seeds (Section VI-D).

Removing the two inspection features from the learned layer changed 23 RF decisions and lowered the hybrid’s accuracy from 0.926 to 0.917 with one C $\rightarrow$ B error. Within the generator the synthetic inspection features thus carried a small amount of signal for the learned layer, whereas they carried none for the rule chain’s grades.

D. Robustness across generator seeds

The pattern held in each of the five additional batches (Table IV, Fig. 5a,b). The floor hybrid introduced no new error relative to the rule chain in any seed, corrected 61 to 68 rule errors per seed (exact McNemar $p < 10^{-18}$ in every seed), and improved accuracy, macro-F1 and balanced accuracy over the paired rule chain by 7.4 ± 0.3 , 7.7 ± 0.4 and 7.1 ± 0.5 percentage points. The RF threshold policy’s accuracy gain over the rule chain reached $p < 0.05$ in only one seed (range $p = 0.017$ to 0.34). In seed 20260932 both components missed the same three C records; the floor hybrid issued B for them (three C $\rightarrow$ B), whereas the C-only floor issued A (three C $\rightarrow$ A), as Proposition 1 predicts. The majority-vote forest’s C recall (0.837 ± 0.013) fell below the rule chain’s (0.842 ± 0.002), so its gain on the original batch did not replicate. The small

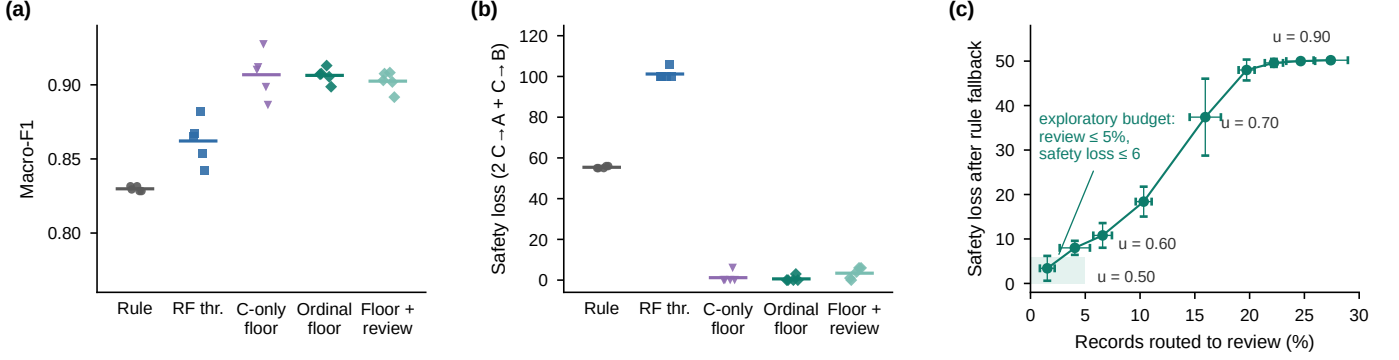


Fig. 5. Five additional generator seeds. (a) Macro-F1 and (b) safety loss L_{safe} per seed (points) and their mean (bar); C-only floor, ordinal floor and floor + review denote RF + C-only floor, RF + ordinal floor and RF + ordinal floor + review ($u = 0.50$). (c) Review-budget sweep for RF + ordinal floor, $u \in \{0.50, 0.55, \dots, 0.90\}$ (mean $\pm$ SD over seeds). Because the offline fallback replaces a routed decision with the rule grade, every routed record can only lose an upward correction, so L_{safe} rises with the review rate. The shaded box marks the exploratory budget of at most 5% review and $L_{\text{safe}} \leq 6$.

between-seed spread shows that the decision-layer behaviour is reproducible within the generator family; all batches share the feature construction, severity mapping and rule logic, so the spread says nothing about transfer to field data.

E. Review budget

Routing low-confidence records to review had a cost that grew with the review rate (Fig. 5c). At $u = 0.50$, $1.5 \pm 0.7\%$ of records were routed, C recall was 0.990 ± 0.008 and L_{safe} was 3.4 ± 2.8 (range 0 to 6); this point met the exploratory budget of at most 5% review and $L_{\text{safe}} \leq 6$ in every seed. At $u = 0.60$ the review rate rose to $6.6 \pm 0.9\%$ and L_{safe} to 10.8 ± 2.8 ; at $u = 0.70$, to $16.0 \pm 1.4\%$ and 37.4 ± 8.6 . On the original batch, $u = 0.50$ routed 7 records and changed no metric, and $u = 0.60$ routed 62 records (7.1%) and produced 12 C $\rightarrow$ B errors (Table III). These costs measure the proxy, not review: the rule grade substitutes for an engineer, so the sweep quantifies the upward corrections that a reviewer would have to recover. A reviewer who returned the reference grade on every routed record would remove these proxy errors; the sweep therefore measures the cost of substituting the rule grade for an unavailable reviewer. The thresholds were selected post hoc within the same generator family and must be fixed before an external evaluation.

F. Comparison with additional baselines

The feature blind spot is not a property of the random forest (Table V). Every learned-only baseline issued at least 50 direct C $\rightarrow$ A decisions on the original batch and between 50.0 and 57.0 on average across the seeds. GBDT was the strongest learned-only model (0.906 accuracy and macro-F1) and still issued 50. The forest assigned $p_C = 0$, and GBDT $p_C \leq 1.1 \times 10^{-6}$, to every through-structure record, so no threshold and no expected-cost rule on these probabilities can recover them: the expected-cost RF, which charges a C $\rightarrow$ A decision five times an A $\rightarrow$ C overestimate, kept all 50. Each learned-only baseline lowered the rule grade on 127 to 179 records, and at most 54 of those overrides were correct.

Two fusion rules that use the rule grade without a floor did not help. The soft vote reproduced the rule chain on the original batch and in four of five seeds (it corrected two rule errors in the fifth), because the one-hot rule grade carries half of the fused probability and prevails unless the forest is certain of a different class. The confidence switch kept all 50 C $\rightarrow$ A errors, because the forest is confident on the blind-spot records. The stacked RF, which receives the rule grade as an input, was the only baseline that outperformed the floor hybrid: 0.994 accuracy, 0.993 macro-F1 and no C-class underestimate. It did so by lowering the rule grade on exactly 50 records, the 25 seepage-mode and 25 mixed-mode A records that the rule chain had escalated to B. All 50 overrides were correct under the benchmark labels.

The floor transferred to every learner. With it, all learners reached zero C $\rightarrow$ A decisions in all six batches. Proposition 1(ii) predicts the price, because the floor inherits each learner’s overestimates: LR + floor and ordinal LR + floor added 47 and 44 errors that the rule chain did not make, SVM + floor 28, expected-cost RF + floor 24 and GBDT + floor one, whereas RF + floor and stacked RF + floor added none in any batch. GBDT + floor was 1.4 percentage points more accurate than RF + floor across the seeds (0.935 ± 0.003 against 0.921 ± 0.004) at the cost of 1.8 ± 1.3 new errors per seed. Stacked RF + floor combined both properties (0.937 ± 0.001 accuracy, no new error).

G. Multi-objective evaluation of the policy space

Of the 4051 policies per batch, 1402 to 1527 satisfied the benchmark constraints and 795 to 825 also satisfied the physical-floor constraint. No learned-only family contained a feasible policy in any batch, and none dominated the rule chain: their hypervolumes ranged from 0.42 to 0.59 against 0.545 ± 0.004 for the rule chain (Fig. 6c). With the ordinal floor, the hypervolume rose to between 0.73 and 0.88 for every learner, and every ordinal family contained a policy that dominates the rule chain (SVM in five of six batches). For the RF, the hypervolume rose from 0.532 ± 0.018 without a floor to 0.804 ± 0.026 with the C-only floor and 0.836 ± 0.010 with the ordinal floor.

TABLE V

COMPARISON BASELINES UNDER THE GROUPED FIVE-FOLD OOF PROTOCOL. COLUMNS UP TO “NEW ERR.” REFER TO THE ORIGINAL 875 RECORDS; THE LAST TWO COLUMNS GIVE MEAN $\pm$ SD OVER THE FIVE ADDITIONAL SEEDS. “BELOW RULE” COUNTS DECISIONS THAT LOWER THE RULE GRADE; “NEW ERR.” COUNTS ERRORS ON RECORDS THAT THE RULE CHAIN GRADES CORRECTLY. LEARNED-ONLY BASELINES USE THE ARGMAX DECISION UNLESS A THRESHOLD IS GIVEN. THE DECLARED FLOOR HYBRID IS SHADED.

	Method	Original batch								Five additional seeds	
		Accuracy	Macro-F1	Bal. acc.	C→A	L_{safe}	A↑	Below rule	New err.	Accuracy	C→A
Physics	Rule chain	.845	.827	.847	0	57	54	0	–	.847 $\pm$.002	0
Learned only	LR	.815	.764	.751	51	102	12	128	128	.828 $\pm$.006	50.8 $\pm$ 0.8
	Ordinal LR	.819	.773	.759	51	102	12	127	124	.823 $\pm$.004	52.8 $\pm$ 1.3
	SVM	.839	.829	.808	63	126	17	142	109	.845 $\pm$.004	56.0 $\pm$ 2.2
	GBDT	.906	.906	.895	50	102	2	130	78	.906 $\pm$.004	50.6 $\pm$ 0.9
	RF, majority vote	.829	.784	.763	51	102	0	179	124	.827 $\pm$.008	57.0 $\pm$ 4.5
	RF, threshold (0.35, 0.20)	.873	.858	.836	50	100	0	154	100	.875 $\pm$.010	50.6 $\pm$ 1.3
	RF, expected cost	.842	.827	.808	50	100	24	154	124	.836 $\pm$.008	50.0 $\pm$ 0.0
Rule-aware fusion	Soft vote	.845	.827	.847	0	57	54	0	0	.848 $\pm$.002	0
	Confidence switch, $u = 0.60$	.858	.839	.823	50	112	0	154	100	.862 $\pm$.009	50.0 $\pm$ 0.0
	Stacked RF	.994	.993	.994	0	0	4	50	0	.995 $\pm$.001	0
With ordinal floor	LR + floor	.855	.815	.814	0	1	58	0	47	.858 $\pm$.002	0
	Ordinal LR + floor	.858	.820	.820	0	1	58	0	44	.859 $\pm$.003	0
	SVM + floor	.870	.849	.857	0	7	71	0	28	.871 $\pm$.005	0
	GBDT + floor	.934	.922	.942	0	0	55	0	1	.935 $\pm$.003	0
	RF, expected cost + floor	.895	.881	.899	0	0	78	0	24	.890 $\pm$.007	0
	RF, threshold + floor	.926	.912	.928	0	0	54	0	0	.921 $\pm$.004	0
	Stacked RF + floor	.937	.926	.947	0	0	54	0	0	.937 $\pm$.001	0

TABLE VI

POLICIES SELECTED BY THE EVALUATION SYSTEM. THE SAME FAMILY WAS SELECTED IN ALL SIX BATCHES, AND THE LEXICOGRAPHIC AND DISTANCE-TO-IDEAL RULES AGREED IN EVERY BATCH. TRANSFER: THE POLICY SELECTED ON THE ORIGINAL BATCH, FROZEN AND EVALUATED ON THE FIVE ADDITIONAL SEEDS (MEAN $\pm$ SD).

	Selected under		Declared reference
	benchmark constraints	+ physical floor	
Policy family	stacked RF	stacked RF	RF
Floor scope (τ_C, τ_B)	none	ordinal	ordinal
	0.50–0.55, 0.35–0.50		(0.35, 0.20)
Accuracy, in batch	.995–.998	.938–.941	.915–.927
Accuracy, transfer	.996 $\pm$.001	.939 $\pm$.001	.921 $\pm$.004
Bal. acc., transfer	.996 $\pm$.002	.948 $\pm$.002	.920 $\pm$.006
Macro-F1, transfer	.995 $\pm$.002	.928 $\pm$.001	.906 $\pm$.005
C→A, transfer	0	0	0
L_{safe} , transfer	0	0	0.6 $\pm$ 1.3
A↑, transfer	3.2 $\pm$ 1.1	53.2 $\pm$ 1.1	53.2 $\pm$ 1.1
Below rule, transfer	50.0 $\pm$ 0.0	0	0

Under the benchmark constraints alone, the Pareto front consisted entirely of stacked RF policies without a floor or with the C-only floor, and the selection rule chose a stacked RF without a floor in all six batches ($\tau_C = 0.50\text{--}0.55$, $\tau_B = 0.35\text{--}0.50$; Table VI). Frozen and transferred to the five seeds, the original-batch selection reached 0.996 ± 0.001 accuracy with no C-class underestimate and lowered the rule grade on 50 records in every seed. With the physical-floor constraint, the front consisted entirely of stacked RF + ordinal floor policies, and the selection kept the thresholds of the unconstrained choice in every batch. The transferred policy reached 0.939 ± 0.001 accuracy, 0.948 ± 0.002 balanced accuracy and no C-class underestimate, 1.8 percentage points above the declared RF + floor. The declared RF + floor was

not on the floor-constrained front in any batch: it is dominated by the variant that uses the rule grade both as an input and as a floor.

The difference between the two selected policies is the measured price of the physical-floor constraint within the generator: 5.7 percentage points of accuracy and 50 additional A-class overestimates. Those 50 decisions are the rule chain’s seepage-gate escalations in the seepage and mixed scenarios. Under the benchmark labels they were all unnecessary. Whether lowering a seepage-gate grade is also safe on a real levee cannot be decided with the present data, and the evaluation system therefore reports the price instead of absorbing it into a single score.

H. Inspection candidates as evaluation states

All nine field candidates that entered manual review were rejected as piping (Fig. 7). Their detector confidences ranged from 0.466 to 0.777. The reviewer attributed five to the thermal contrast at water–land boundaries, two to vegetation and shadow, one to the thermal inertia and shadow of a building edge, and one to wet ground or lens-edge vignetting. With 0 confirmations in 9 reviews, the exact one-sided 95% upper bound on candidate precision is 0.28: the data exclude a high-precision detector, but they do not show that the detector never finds piping. The 14 boxes on the 13 artificial-piping images had higher confidences (0.670 to 0.930), but without missed-detection records they establish no recall or field transfer.

The implication for the inspection indicator is direct. A candidate may create a review task, but it cannot enter the grade as a confirmed piping event until time, location and manual status are closed, and an image that was not inspected remains unobserved rather than “no anomaly”. The two highest-confidence field candidates (0.777 and 0.767)

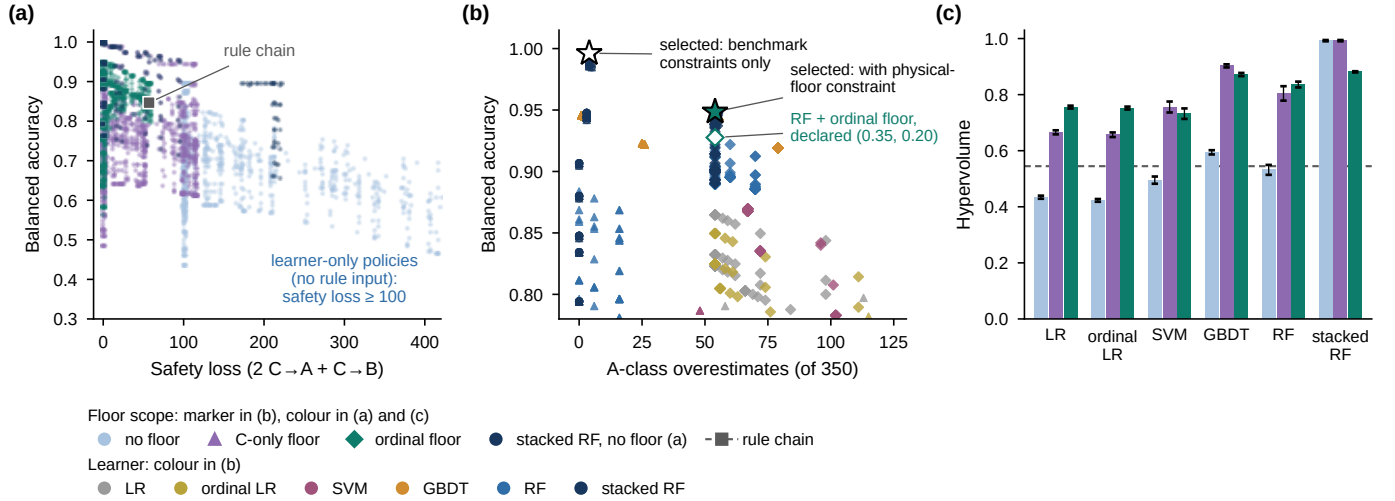


Fig. 6. Multi-objective evaluation of the 4051 candidate policies. (a) Balanced accuracy against safety loss for all policies on the original batch; colour gives the floor scope, and stacked RF policies without a floor are shown separately because they use the rule grade as an input. (b) Policies that satisfy the benchmark constraints ($N(C \rightarrow A) = 0$, $R_C \geq 0.95$), plotted against the A-class overestimate burden; marker gives the floor scope and colour the learner; policies with balanced accuracy below 0.78 lie outside the panel. Stars mark the policies selected without and with the physical-floor constraint; the open diamond marks the declared RF + ordinal floor. (c) Hypervolume of each learner and floor scope, Eq. (10), mean $\pm$ SD over the original batch and five seeds; the dashed line is the rule chain.

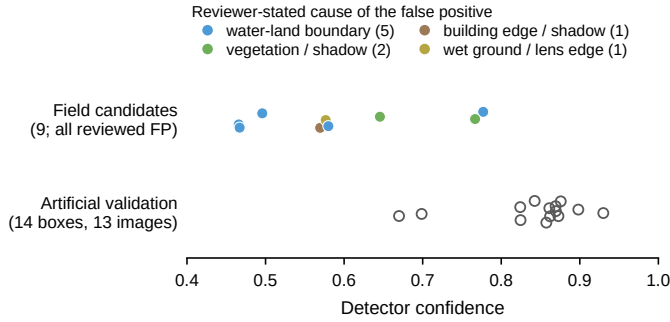


Fig. 7. Detector confidence of the nine reviewed field candidates, coloured by the reviewer-stated cause of the false positive, and of the 14 boxes on artificial-piping images (no review status or missed-detection record). Vertical position within a row is jitter for visibility.

were a water edge and vegetation, so a confidence threshold alone would not have separated them from genuine seepage.

I. Engineering-case provenance trace

The K45+500 delivery returned a C grade with a seepage-dominant mode and a score of 24.35 (Table VII). The provenance fields exposed five issues that prevent its use as validation: the water level is a representative high-water scenario rather than a measured 2017 level; 15 of the 29 indicators were reported as missing; four indicators conflict between the delivered indicator table and the result file; the evaluation module used for the case differs from the benchmark version and adds a rule that sets the inspection indicator to its worst value when piping is confirmed; and the confirmed historical piping event is part of the input. The trace therefore demonstrates that the interface carries the information needed to detect these problems; it does not test predictive skill.

TABLE VII
K45+500 DELIVERY CASE: VALUES AND PROVENANCE STATUS.

Item	Delivered value	Provenance status
Upstream water level	35.90 m (crest 36.30 m)	representative scenario, not a measured 2017 level
Controlling layer	silty clay, $i = 0.865$, $i_{cr} = 0.827$, ratio 1.045	numerical output; extraction location and material retained; C1 gate fired
Downstream slope	FoS 1.155 (Bishop simplified)	below the 1.35 allowable value; D1 gate fired
Scour depth	0.40 m	D3 gate fired
Score and grade	24.35; C, seepage-dominant	delivery excerpt; no independent re-run
Completeness	15 of 29 indicators missing	reported as a missing-item list
Consistency	4 table–result conflicts	unresolved (A8, A9, B2, D5)
Code version	module differs from benchmark	adds “confirmed piping → worst C3”; history enters the input

VII. DISCUSSION

The safety gain in this paper comes from the composition, not from the learner. A forest trained on the same physical channels improved aggregate metrics only modestly and, on every batch, converted its critical-class misses into direct downgrades; the ordinal floor removed those downgrades and kept most of the learned corrections. Proposition 1 explains why this worked here and why it need not work elsewhere: the floor is valuable exactly when the two components err on different records and the learned layer rarely overestimates. In the benchmark, 125 of the rule chain’s 136 errors fell in the safe-reference, overtopping, seepage and mixed modes, 100 of the forest’s 111 errors fell in the through-structure and overtopping modes, and the two never missed the same C

record.

This positions the floor between two existing strategies. Theory-guided learning places physical knowledge in the learner’s inputs, loss or architecture [11], and interpretable models make the learner itself auditable [12]. The floor instead places physical knowledge in the *decision layer* as an ordinal constraint, which leaves the learner unrestricted but bounds the direction of its influence. It is simpler than Bayesian updating of failure probabilities [7], [8] and does not replace it; the two answer different questions. Unlike a reject option [14], [15], the floor never abstains; in our sweep, abstention with a rule-based fallback only removed upward corrections, so the value of review lies in information that the offline proxy cannot represent.

Two rival explanations bound the size of the gain. First, the rule chain’s hard gates are aligned with the generator’s severity construction, so the reference labels partly encode the rule logic; part of the hybrid’s advantage over a pure learner is therefore built into the benchmark. Second, the forest’s C→A errors stem from a feature set that omits structure indicators. The stacked RF confirms this reading: given the rule grade, which encodes the structure gates, the forest recovered every through-structure critical case. Neither explanation removes the protective role of the floor, which does not depend on knowing where the learner’s blind spots are. That property matters for field deployment, where blind spots are unknown in advance.

The baselines and the policy-space evaluation sharpen the role of the floor. The blind spot belongs to the features, not to the forest: logistic, ordinal, support-vector and boosted models missed the same critical cases, and neither soft voting nor confidence switching removed the misses. The only composition that outperformed the floor, the stacked RF, did so by lowering 50 rule grades. Within the generator those overrides were correct; on a levee they would overrule seepage gates on the strength of regularities learned from synthetic labels. The evaluation system therefore reports two operating points instead of one: a floor-admissible policy that is deployable under the present evidence, and a benchmark-optimal policy whose additional 5.7 points depend on field confirmation of learned downward overrides. Among the floor-admissible learners the forest had a specific advantage, because it never overestimated and therefore added no error to the rule chain; supplying the rule grade as an input kept that property and raised accuracy to 0.939 ± 0.001 . Making the constraints, the selection rule and the price of each constraint explicit is the practical contribution of the Pareto layer: it replaces a trade-off hidden in a single score with a declared and auditable choice.

The inspection results lead to the same design principle at the evidence layer. A detector can be optimized for candidate generation, but the evaluation should consume reviewed event objects with a location, a time window, a confidence and a status. Retaining rejected candidates is useful for retraining and flight planning [29]; promoting them to confirmed piping would have raised the grade of a reach on the basis of water edges and vegetation. The provenance graph keeps that distinction in the data model rather than in the visualization layer.

The claims are bounded in five ways. (i) All quantitative comparisons use one synthetic generator; the five additional seeds test realization variability, not transfer, and the reference labels share the rule logic. (ii) The C→A guarantee is conditional on the rule chain: in the field, a rule chain that misses a critical case entirely would pass that error through the floor, and the hybrid inherits every rule overestimate, which in the benchmark meant 54 unnecessary escalations. (iii) The review fallback is a proxy, so the reported review costs are not measurements of engineering review. (iv) The field evidence consists of nine reviewed candidates and one non-blind engineering trace; it supports the evidence-state design but no detection or prediction metric. (v) The objectives, the 2:1 weighting in L_{safe} , the constraint thresholds and the lexicographic order are declared rather than calibrated (Table II); the transfer test guards against selecting on one batch, not against selecting within one generator.

Each boundary points to a discriminating test. Independent field labels, timed and registered against inspection and water-level records, would test whether the components remain complementary outside the generator, which Proposition 1 identifies as the condition that matters. Retraining the learner with structure indicators would separate the feature blind spot from the protective effect of the floor. Field labels on units where the stacked RF would lower a seepage-gate grade would decide whether the physical-floor constraint can be relaxed. Calibrating the forest’s probabilities and recording actual reviewer decisions would turn the review gate from a proxy into a measured operating point.

VIII. CONCLUSION

An ordinal physical floor that lets a learned layer raise but never lower a physics-based grade improved a levee seepage rule chain without adding a single error relative to that chain in six controlled batches, and removed the direct critical-to-safe downgrades that no threshold or moderate cost weighting of a random forest could avoid. The gain follows from a simple property: the composition underestimates only where both components underestimate, so it helps when their errors are complementary and protects against direct downgrades only to the extent that the rule chain does. A multi-objective evaluation of 4051 policies per batch made the remaining trade-off explicit: the benchmark-optimal stacked RF gained 5.7 points of accuracy only by lowering 50 rule grades, whereas the best floor-admissible policy, the same forest with the ordinal floor, transferred across five generator seeds at 0.939 accuracy without a critical-class underestimate. Treating inspection detections as candidates until reviewed, and engineering deliveries as provenance traces, kept unconfirmed and inconsistent evidence out of the grade. Field labels with independent timing and registration are the next requirement for testing whether the complementarity, and hence the gain, survives outside the synthetic generator.

DATA AND CODE AVAILABILITY

The de-identified repository contains the benchmark and its generator (data/875_benchmark), the multi-seed batches and

analysis scripts (experiments/analysis), the inspection audit tables (inspection/audit) and this manuscript. The statistics, baselines, policy-space evaluation and figures of this paper, including per-record predictions and the cloud of 4051 policies, are produced by the scripts in experiments/analysis/paper_revision, which reuse the unchanged model code. Raw field images were released with metadata removed where possible; locations are not reported.

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